

Analyzing the Effect of Network Width on Graph Transformers

Progress review 1 (Явц 1) — weeks 1–5

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Diploma research

How does a graph transformer's width — its embedding dimension m — decide what it can learn, and how must width grow with graph size n ?

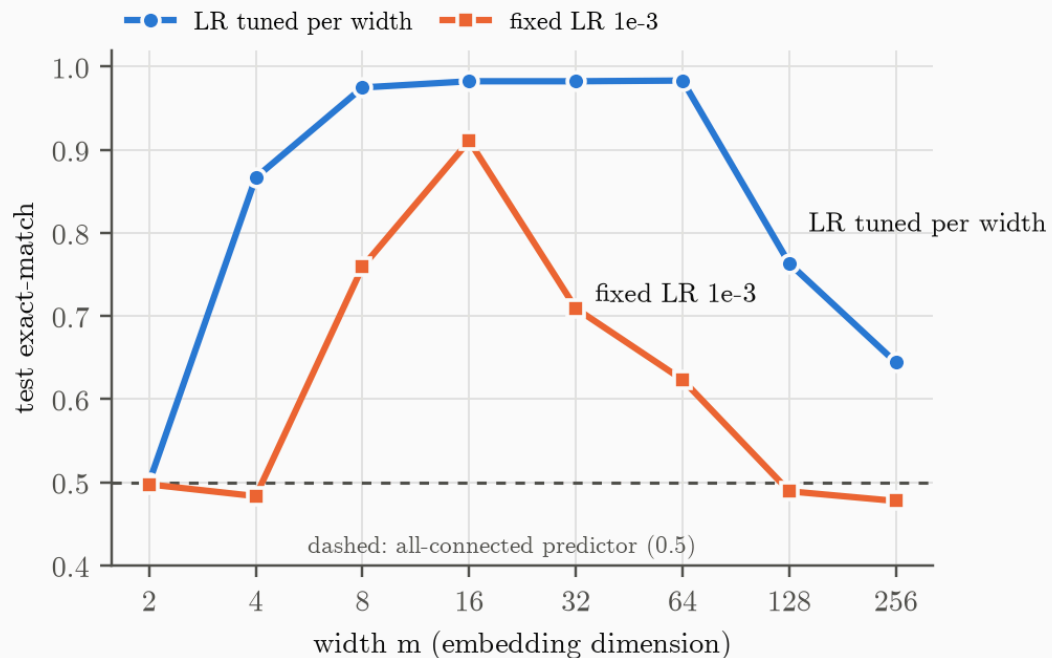
- **Theory** gives width bounds for *representing* graph algorithms: one attention layer needs width $\approx$ (what each token must gather) — q-sparse averaging, Sanford et al. 2023; connectivity at depth $\log n$ needs only n^ϵ width (Sanford et al. 2024).
- **Nobody has measured** the width a transformer needs to *learn* a graph task, how it scales with n , or how it trades against depth.

This thesis measures the **critical width** m^* — the smallest width that learns the task — and how it moves with n , depth, data and training.

- **Task:** connectivity matrix — for every pair of nodes, are they in the same component? (dense supervision; a single yes/no per graph does not train)
- **Model:** transformer encoder, one token per node (its adjacency row), pairwise read-out — task, tokens and read-out from Ye et al. (2026)
- **Width** m = embedding dimension; the whole model scales with it: $m/8$ attention heads of size 8, feed-forward layer $4m$
- **Metric:** pair accuracy (fraction of pairs right); trivial “all connected” ≈ 0.75
- **Critical width m^* :** where accuracy crosses 0.95, per seed
- **Protocol:** choices on validation, results on test; 2–3 seeds
- **Compute:** sharded sweeps on Kaggle GPUs; ≈ 770 training runs so far

Part 1 — Measuring width honestly

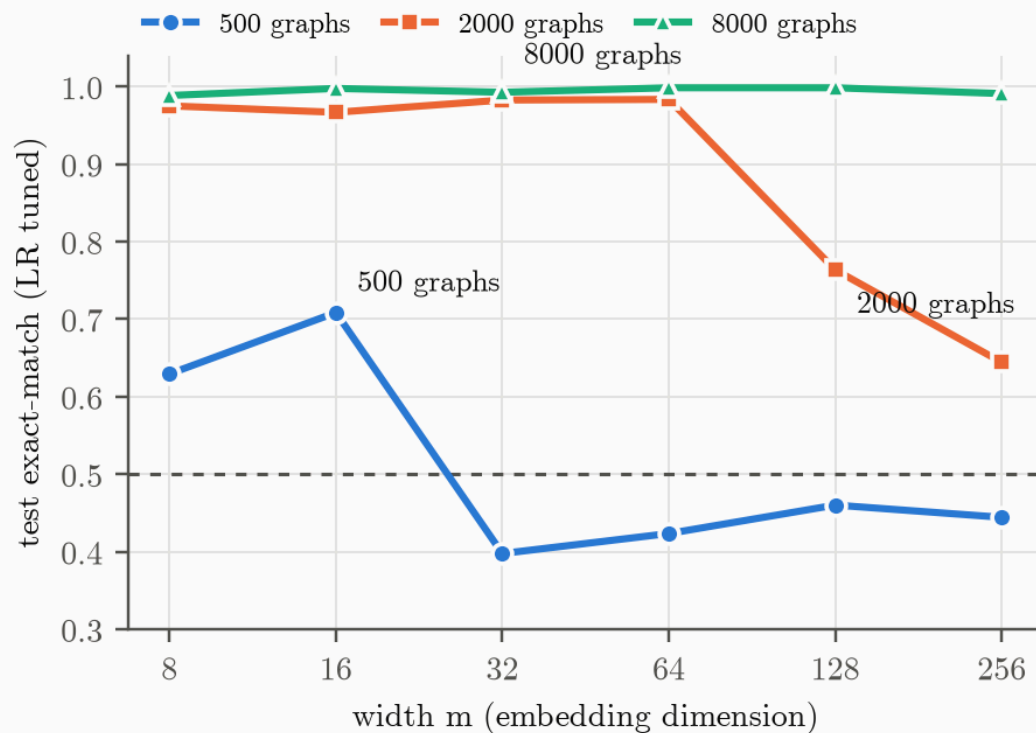
Lesson 1: a fixed learning rate fakes a width effect



- One fixed rate: wide models look up to **0.4 worse**
- Tuned per width: widths 8–64 all ≈ 0.97 – 0.98
- The best rate **falls as width grows**

$n = 24$, depth 2, 2000 graphs, 3 seeds (Q1/Q1b, 192 runs)

Lesson 2: width has a floor and a data ceiling



- **Floor:** below $\approx 4-8$ the model cannot even fit
- **Ceiling:** past it, extra width memorizes
- More data moves the ceiling: at 8000 graphs every $m \geq 16$ reaches ≥ 0.99

same steps for every data size (Q1c, 216 runs)

Lesson 3: the benchmark was leaking

The first dataset could be solved without tracing a single path:

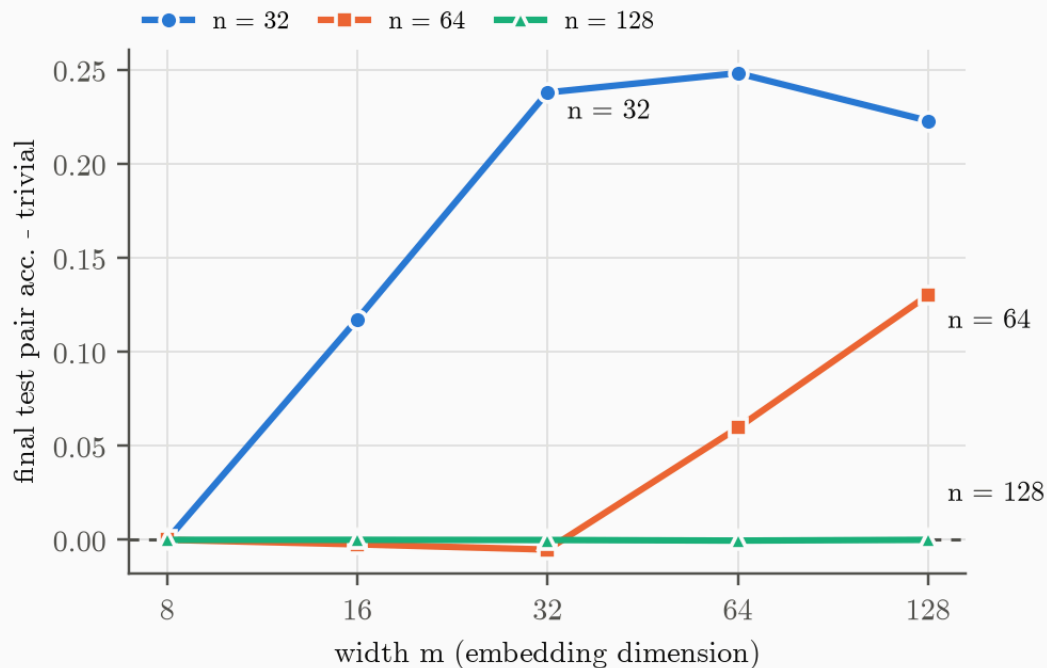
shortcut	evidence
node indices reveal the components	blobs always nodes $[0, n_a)$ and $[n_a, n)$ — 200 / 200 graphs
graph statistics reveal the label	degree + short-cycle counts predict it with 0.68–0.90 accuracy
small graphs are local	a 6-hop check is exact on 60–100 % of graphs

Fix — the swap generator: the two classes differ only by swapping two far-apart edges (the two-cycles-vs-one “cycle task” of Abbe et al., NeurIPS 2024, plus chords and a dense target); node labels shuffled. Audit: statistics **at chance (0.49–0.53)**, 4-hop checks **score 0**, diameter grows $\approx \log n$.

On the leaky data width 8 sufficed at $n = 24$ — the shortcuts hid the width requirement.

Part 2 — How width must grow

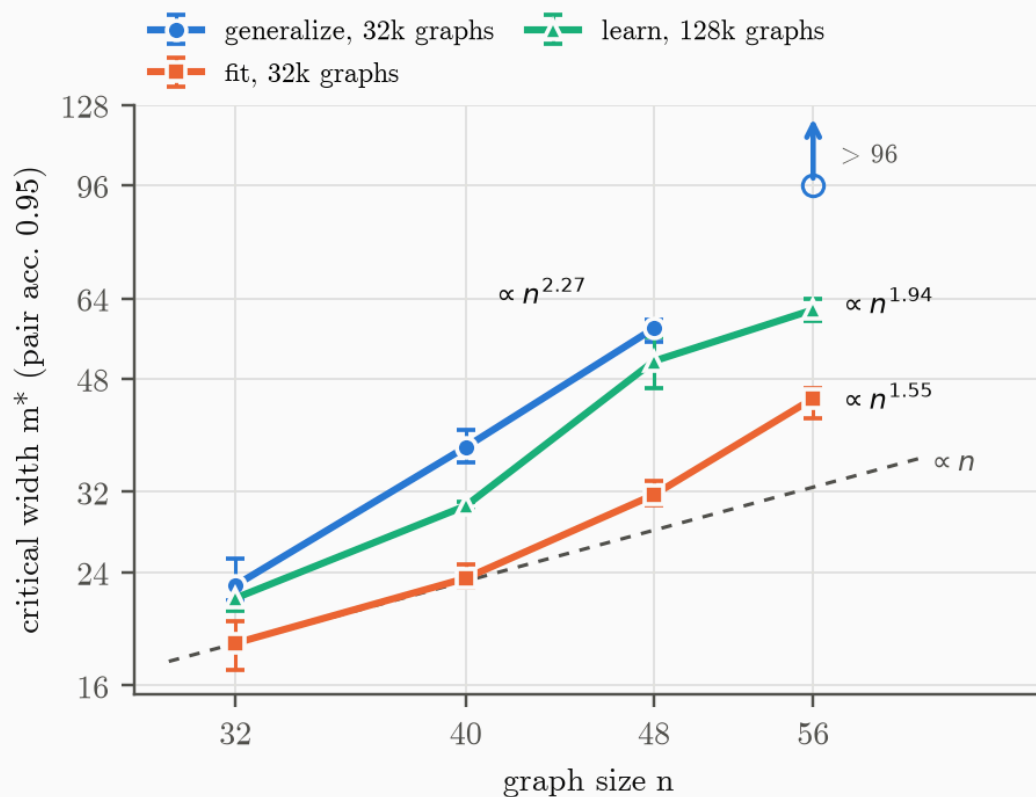
Width decides whether learning starts at all



- $n = 32$: learns from $m \approx 16$
- $n = 64$: only $m \geq 64$ leave the trivial predictor
- $n = 128$: **nothing** up to $m = 128$ learns

depth $\lceil \log_2 n \rceil$, 32 000 graphs, 12 000 steps

Main result: the critical width grows $\approx n^2$



n	m^*	m^*/n
32	21.9	0.68
40	30.2	0.76
48	51.1	1.07
56	57.9	1.03

$$m^* \propto n^{1.85} \quad (90\% \text{ range } 1.68\text{--}2.03)$$

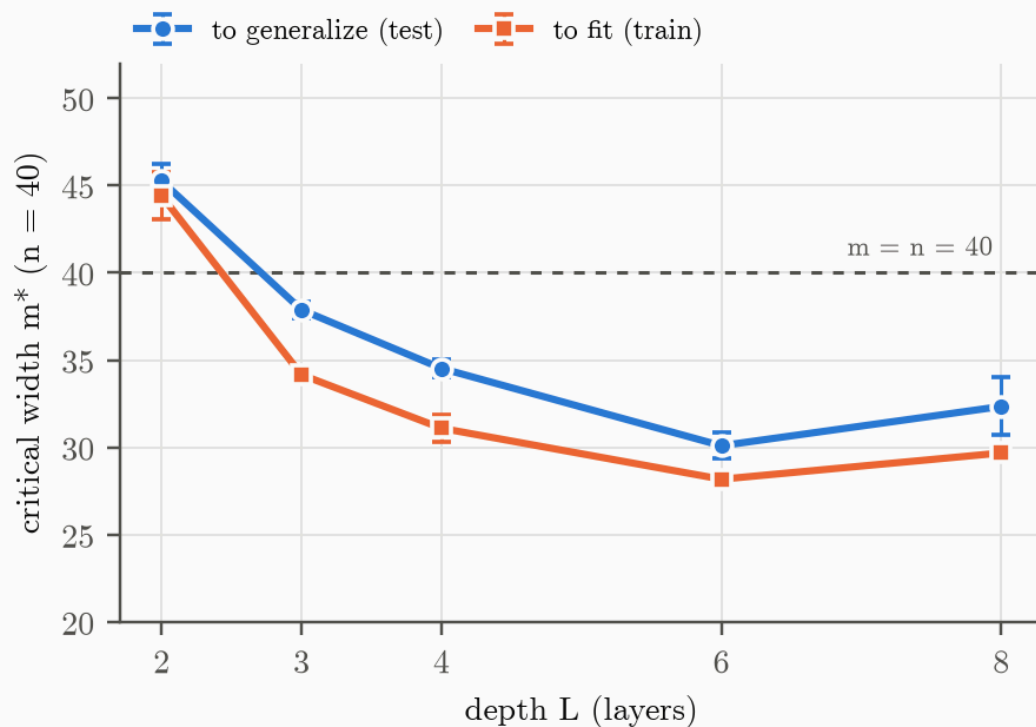
depth 6, 128 000 graphs, 24 000 steps; table: rate tuned at $n = 40 / 56$
(green line in the chart: one rate, $n^{1.94}$)

The result survives five checks

could it be...	test	outcome
too little data?	32k $\rightarrow$ 128k graphs	gap closes, slope stays ≈ 2
too little training?	2 $\times$ the steps	critical width unchanged
the learning rate?	second rate, tuned per width	slope 1.94 $\rightarrow$ 1.85
the input encoding?	fixed 96-wide random node IDs	still $\approx n^2$ (2.03)
noise?	2–3 seeds, per-seed crossings	seeds within ± 3–5

Over $n = 32$ –56, the width a transformer needs to **learn** connectivity grows roughly as n^2 — while theory says this depth needs far less width to **represent** it.

Depth trades for width only mildly



- **2 layers learn** at $m \approx 1.1n$ — on graphs of diameter ≈ 10 , so the model does not trace paths hop by hop
- $2 \rightarrow 6$ layers saves only $\approx \frac{1}{3}$ of the width; $8 = 6$
- Matches Yehudai et al. 2025: linear width $\Rightarrow$ constant depth

$n = 40, 60$ runs

What it means



1. **A protocol for measuring width.** Learning rate, data and training budget each interact with width enough to fake or hide an effect; we control all three.
2. **An audited benchmark.** Every heuristic we could find sits at or below chance on swap; the leaky data had hidden the width requirement.
3. **A measured scaling law.** The critical width to learn connectivity grows $\approx n^{1.9}$ over $n = 32\text{--}56$, robust to five confounds.
4. **Representation $\neq$ learning.** Theory: depth $\log n$ needs sublinear width to represent connectivity. Measured: learning needs $\approx n^2$. Depth helps only mildly.

- Graph sizes span less than a factor of two ($n = 32\text{--}56$); larger n did not learn at any affordable width
- Two seeds for most scaling points; local slopes vary ($\approx 1.2\text{--}2.9$)
- One task (connectivity) and one architecture family so far
- swap diameter grows with $\log n$: path length and node count still rise together
- Compute: Kaggle's 30 GPU-hours per week caps each sweep

item	question	when
Q7c	does the depth-2 critical width grow linearly or $\approx n^2$?	week 6
Q7b	fix n , vary diameter: is path length or node count driving the growth?	week 6
Q4	Graphormer-style distance bias — on a task it does not give away	weeks 6–7
Q2	task hierarchy: retrieval < connectivity < shortest path	week 7